

- 4 An object is thrown vertically upwards in air in which the air resistance is not to be neglected.

If the times of flight for the upward motion t_u and the time of flight to return to the same level t_d are compared, then

- A $t_d > t_u$, because the object moves faster on its downward flight and therefore the air resistance is greater.
- B $t_d = t_u$, because the effect of the air resistance is the same whether the object is moving upwards or downwards.
- C $t_d < t_u$, because at any given speed the net force when the object is moving downwards is greater than the net force when it is moving upwards.
- D $t_d > t_u$, because at a given speed the net force when the object is moving downwards is smaller than the net force when it is moving upwards.